

MH1301 Discrete Mathematics

Handout 9: Graph Theory (3): Graph Coloring, Planar Graphs

Here is the third handout on graph theory. We study the coloring of graphs, to be more precise, the coloring of the vertices of graphs so that adjacent vertices have different colors. We find it appropriate to introduce Bipartite Graphs in this handout as they are the “2-colorable” graphs. We also study planar graphs, which are graphs that can be drawn in a 2D plane without any of the edges crossing.

- Bipartite Graphs,
- Graph Coloring,
- Planar Graphs.

Notes on Simple Paths

Recall from Definition 2 (Section 10.4):

A **simple path** is a path that contains no edge more than once.

We will use this definition in this course.

Alternative Definition

A **simple path** is a path that contains no vertices more than once.

Other books/authors may use this alternative definition.

Conventions about graphs and terminologies in this course

- (1) A graph $G = (V, E)$ has $|V| \geq 1$. Unless otherwise specified, we assume graph is *simple*.
- (2) A path has length at least 1.
- (3) A circuit has length at least 1 (loop), and at least 2 for simple graph (which has no loop).

Coloring Graphs

Definition 1 (Section 10.8)

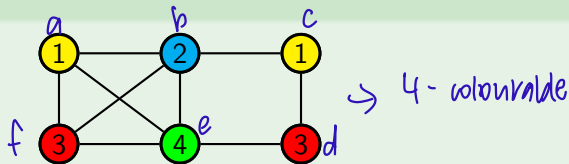
A k -vertex-coloring of a graph $G = (V, E)$ is a function

$$f : V \mapsto \{1, \dots, k\}$$

such that for all $u, v \in V$ with $(u, v) \in E$, we have $f(u) \neq f(v)$.

Intuitively, a k -vertex-coloring of a graph is a way to color each vertex one of k different colors, such that no two adjacent vertices have the same color.

Example



Chromatic Number

Definition 2 (Section 10.8)

- A graph G is k -colorable if a k -vertex-coloring of G exists.
- The chromatic number of a graph G is the smallest k for which G is k -colorable.
- The chromatic number is denoted by $\chi(G)$.

Examples

all vertices are connected with edges

- $\chi(K_n) = n$.
- $\chi(C_n) = 2$ for even n . *cycles*
- $\chi(C_n) = 3$ for odd n .
- $\chi(G) = 1$ if and only if the graph G has no edges.

However, in general computing $\chi(G)$ is a hard problem.

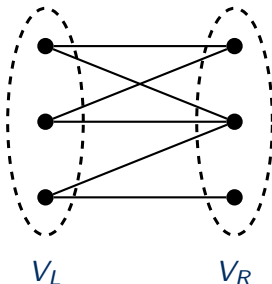
Bipartite Graphs

Definition 6 (Section 10.2)

won't connect to those in same set

A (simple) graph $G = (V, E)$ is called **bipartite** if its vertex set V can be partitioned into two disjoint sets V_L and V_R such that every edge in the graph connects a vertex in V_L and a vertex in V_R (so that no edge in G connects either two vertices in V_L or two vertices in V_R).

We call the pair (V_L, V_R) a **bipartition** of the vertex set V of G .



Theorem 4 (Section 10.2)

A graph $G = (V, E)$ is **bipartite** if and only if $\chi(G) \leq 2$.

If $\chi(G) = 1$, then G consist of isolated vertices only...

Proof (“if”): Suppose $\chi(G) = 2$. Let V_L be the set of vertices assigned one color and let V_R be the set of vertices assigned another color. Then clearly

$$V_L \cap V_R = \emptyset \quad \text{and} \quad V = V_L \cup V_R.$$

Clearly there are no edges in G connecting two vertices in V_L nor two vertices in V_R , since vertices in V_L are of the same color and likewise for vertices in V_R .

(“only if”): Suppose $G = (V, E)$ is bipartite, i.e., there is a bipartition (V_L, V_R) where no edge in G connects either two vertices in V_L or two vertices in V_R .

If $E \neq \emptyset$, then $\chi(G) = 2$

If $E = \emptyset$, then $\chi(G) = 1$

It suffices to color vertices in V_L one color, and those vertices in V_R another color.

Bipartite Characterization

Theorem (Ex. 10.4.49)

A graph G is bipartite if and only if there is no subgraph of G that is an odd cycle.

$$\neg q \Rightarrow \neg p$$

$\Rightarrow$ • An odd cycle is not bipartite, and adding more edges and vertices does not make the resulting graph bipartite.

$\downarrow$ length of cycle is odd

$\Leftarrow$ • The other direction is also quite easy, but we skip the proof.

$$\neg p \Rightarrow \neg q$$

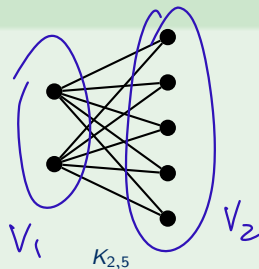
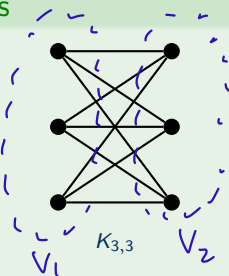
Special Graphs

Complete Bipartite Graph (Example 13, Section 10.2)

The **complete bipartite graph** $K_{m,n}$ has $m + n$ vertices: $V = V_1 \cup V_2$, $|V_1| = m$, $|V_2| = n$, $E = V_1 \times V_2$ such that

- there are **no edges within V_1 or V_2** , and
- for each $u \in V_1$ and $v \in V_2$, there is an edge between u and v .

Examples

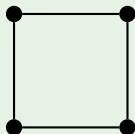


Planar Graphs

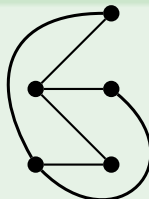
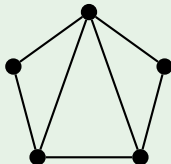
Definition 1 (Section 10.7)

A graph is called a **planar graph** if there is some way to draw it in a 2D plane without any of the edges crossing, where a crossing of edges is the intersection of the lines at a point other than their endpoints. Such a drawing is called a **planar representation** of the graph.

Examples



C_4



$K_{2,3}$

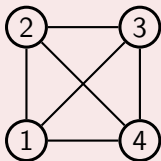
Planar Graphs

Wait a minute, didn't you say at the beginning that the location of vertices and edges in the drawing is **not** part of the graph?

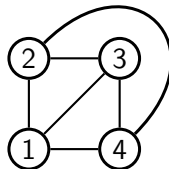
A graph is called a **planar graph** if **there is some way** to draw it in a 2D plane without any of the edges crossing.

Question

Is K_4 shown below a planar graph?



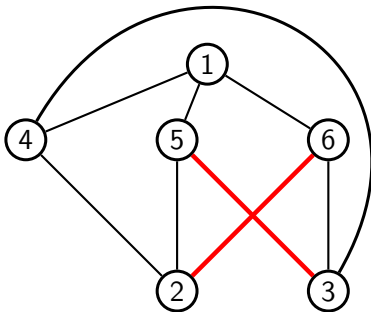
Solution: Yes! Because it can be drawn as



Non-Planar Graph

Example 3 (Section 10.7)

Is $K_{3,3}$ a planar graph?



$K_{3,3}$ is not a planar graph.

Checking Planarity

Question

How to check whether a graph is a planar graph?

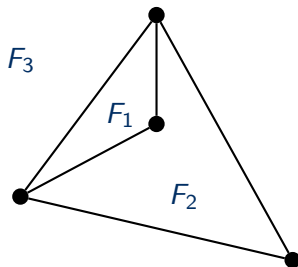
A fun game:

<http://www.nkhem.com/planarity-knot/>

Regions

Region

A planar representation of a graph splits the plane into **regions** (or **faces**).



The region F_3 is called the **infinite region**.

Euler's Formula

Euler's Formula (Theorem 1, Section 10.7)

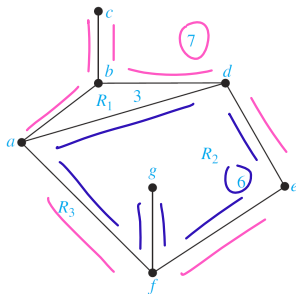
Let G be a connected planar simple graph with m edges and n vertices. Let r be the number of regions in a planar representation of G . Then

$$r = m - n + 2.$$

- We skip the proof (which uses induction). *(on number of edges)*
- For interested students: a fantastic video on Euler's Formula:
<https://youtu.be/-90Uyo8NFZg>

Euler's Formula

For a planar graph drawn without edges crossing, the number of edges bordering a particular region R , denoted by $\deg(R)$, is called the **degree** of the region. When an edge occurs twice on the boundary, it contributes two to the degree.



$$\begin{aligned} \bullet & \deg(R) = 0 \\ \bullet & \deg(R) = 2 \end{aligned}$$

$$\begin{aligned} \bullet & \deg(R) \geq 3 \\ \bullet & \deg(R) = 3 \\ \bullet & \rightarrow \deg(R) = 4 \end{aligned}$$

Proposition (Sum of the Degrees of Regions)

In any planar simple graph, the sum of the degrees of all regions is equal to twice the number of edges.

Corollaries

Corollary 1 (Section 10.7)

If G is a connected planar simple graph with m edges and n vertices, where $n \geq 3$, then

$$m \leq 3n - 6.$$

Proof: Every region of that graph has at least 3 bounding edges, and so

$$2m = \sum_R \deg(R) \geq 3r \text{ or } r \leq \frac{2}{3}m.$$

From the Euler's formula we deduce that

$$2 = n + r - m \leq n + \frac{2}{3}m - m = \frac{3n - m}{3},$$

in other words, $3n - m \geq 6$.



Non-planarity if K_5

Example 5 (Section 10.7)

Prove that the complete graph K_5 is not planar.

Proof: The graph K_5 has $n = 5$ vertices and $m = 10$ edges.

If K_5 would be planar, then by Corollary 1 it follows that $m \leq 3n - 6$, in other words, $10 \leq 9$, which is a contradiction. $\square$

- Planar graphs cannot have too many edges.
- K_5 has too many edges to be planar.

Planar Graph and Cycles

Corollary 3 (Section 10.7)

If G is a connected simple planar graph that has at least 3 vertices and no cycles of length 3, then

$$m \leq 2n - 4.$$

Proof: Every region of G has at least 4 edges, hence $2m \geq 4r$. Then, from the Euler's formula we get

$$2m = \sum_R \deg(R) \geq 4r \quad \text{---}$$
$$2 = n + r - m \leq n + \frac{1}{2}m - m = n - \frac{1}{2}m$$

in other words, $2n - m \geq 4$. □

Non-planarity of $K_{3,3}$

Example 6 (Section 10.7)

Prove that the complete bipartite graph $K_{3,3}$ is not planar.

Proof: The graph $K_{3,3}$ has $n = 6$ vertices and $m = 9$ edges. This graph cannot contain cycles of length 3. $\rightarrow$ bipartite $\chi(K_{3,3}) = 2$

If $K_{3,3}$ would be planar, then by Corollary 3 it follows that $m \leq 2n - 4$, in other words, $9 \leq 8$, which is a contradiction. □

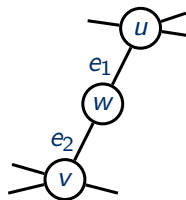
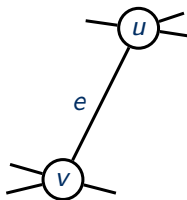
Also checkout this video for a fun explanation!

<https://youtu.be/VvCytJvd4H0>

Homeomorphic Graphs (Optional)

Elementary subdivision operation

- For a graph G , if we remove an edge $e = \{u, v\}$, add a new vertex w , and then insert the edges $e_1 = \{u, w\}$, $e_2 = \{w, v\}$, then this is called an **elementary subdivision**.
- Two graphs G_1 and G_2 are called **homeomorphic** if they can be obtained from the same graph by a sequence of elementary subdivisions.



Kuratowski's Theorem (Optional)

Theorem 2 (Section 10.7)

A graph is non-planar if and only if it contains a subgraph that is homeomorphic to either K_5 or $K_{3,3}$.

In other words, a graph G is not planar if and only if it “contains” K_5 or $K_{3,3}$ as follows:

- the vertices of K_5 or $K_{3,3}$ are vertices of G ,
- the edges of K_5 or $K_{3,3}$ are simple paths of G with no common vertices (except start and end vertices).

Map Coloring Problem

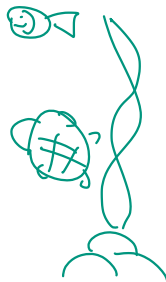
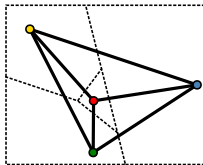
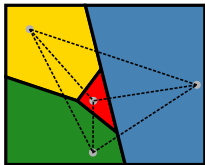


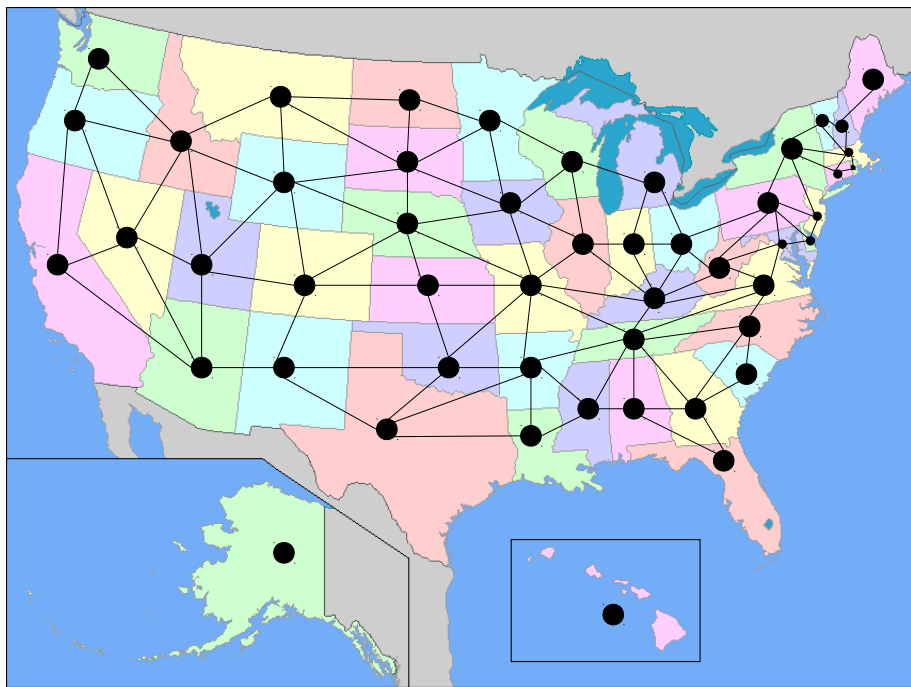
Given any separation of a plane into contiguous regions, producing a figure called a **map**. Two regions are called **adjacent** if they share a common boundary that is not a corner.

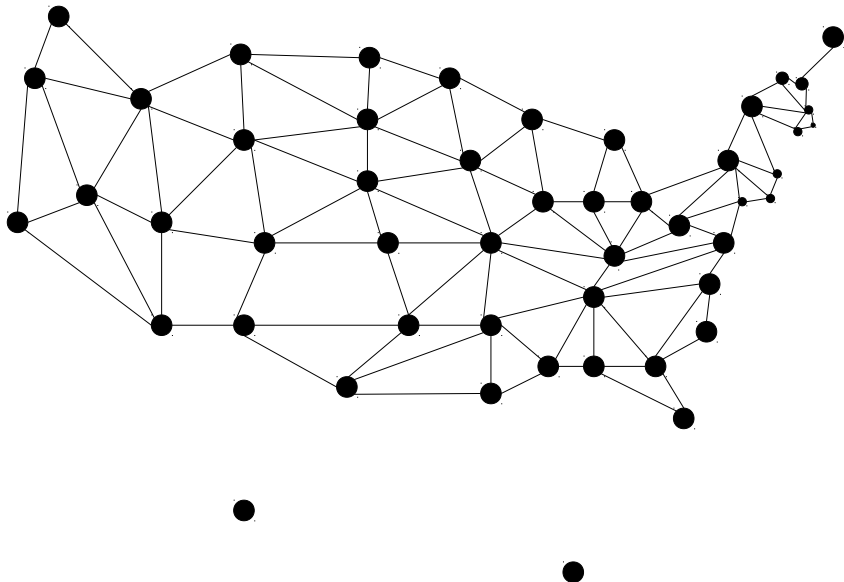
Question

How many colors are required to color the regions of the map so that no two adjacent regions have the same color?

The problem can be easily reduced to the problem of **planar graph coloring**.







Four-Color Theorem

Theorem 1 (Section 10.8)

Every simple planar graph is 4-colorable.

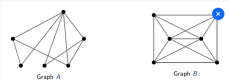
- The problem was originally posed in 1852.
- The first proof by Alfred Kempe, 1879 (an error was found in this argument in 1890).
- Proofs by combination of computer search and theoretical reasoning:
 - ▶ Kenneth Appel, Wolfgang Haken, 1976,
 - ▶ Neil Robertson, Daniel Sanders, Paul Seymour, Robin Thomas, 1997,
 - ▶ Georges Gonthier, 2002.
- Not all mathematicians accept these proofs.

Philosophical Question: Is a theorem true if no human has ever read the proof?

Reading Assignment

- Rosen (7th edition):
 - ▶ Chapter 10.7. Planarity
 - ▶ Chapter 10.8. Coloring

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Graph A

Graph B

Refer to the diagram and answer the questions below.

Waiting for next clap

(a) [T for true, F for false] Graph A is planar. Answer: **T**

(b) [T for true, F for false] Graph B does not contain cycles of length 3.
Answer: **F**

(c) Number of edges in Graph $B = 13$

(d) Number of vertices in Graph $B = 6$

(e) [T for true, F for false] Graph B is planar. Answer: **F**